

Final Report

To accompany their digital intervention, student groups write the Final report that documents the key steps in the intervention development, design and implementation process. The Final Report is grounded in an extensive literature review and desk research, incorporating relevant social science theories to contextualize the project within its broader academic and policy frameworks. This report outlines the datasets and methodologies applied and includes a critical reflection on the limitations and potential biases of the chosen approaches and the results obtained. The Final report also highlights the key findings of the project, offering insights into its relevance to stakeholders, including the Project partner. It evaluates the potential societal and policy impacts of the intervention, addressing its significance and contributions to the identified challenge.

1. Title Page

- Project title
- Group information

2. Executive Summary 400-600 words

3. Introduction and Background 900-1200 words - Meng

Urban heat is no longer a seasonal issue but a structural public health crisis. In Amsterdam, this challenge is defined by the Urban Heat Island (UHI) effect, where the city's dense architecture and impervious surfaces cause temperature differentials of up to 11.6°C compared to surrounding rural areas.

The core challenge addressed by our project is the Information-Action Gap. While the municipality possesses high-resolution climatic data (e.g., the Amsterwarm project), this data is often static and "technical. It does not translate into decision-ready information for a caregiver standing on a hot street corner with an infant or an elderly person. Currently, cooling resources exist as fragmented datasets. There is no unified, real-time interface that integrates indoor climate-controlled sanctuaries with outdoor shading, making the city's cooling infrastructure invisible to those who need it most.

In Computational Social Science (CSS), vulnerability is defined as a dynamic intersection of Exposure, Sensitivity, and Adaptive Capacity. This framework moves beyond simple meteorology to analyze how heat risk emerges when high Land Surface Temperatures (Exposure) collide with biological vulnerabilities like age or illness (Sensitivity). The critical part

is Adaptive Capacity: while affluent residents utilize private cooling like insulation or AC, vulnerable populations rely on public adaptive capacity, such as municipal libraries and shaded squares.

By mapping these dimensions, our project identifies the Resilience Gap, the systemic mismatch between where heat is most intense and where cooling resources are actually accessible. This transforms the "Cool Map" from a technical tool into an Environmental Justice intervention. It specifically targets the "urban green inequality" that leaves low-income or heat-sensitive residents without the means to adapt, using data to bridge the gap between where the heat is and where the help is.

Developing a digital intervention requires a complex ecosystem of stakeholders, each stakeholder will have impact on the project's design and success. At the center of this ecosystem are the primary users, including caregivers and heat-sensitive residents, whose interests prioritize safety, reliability, and access to cooling information. Despite having a high moral claim to the city's resources, this group often holds the least structural power and is losing access to critical information exactly when environmental stress is highest. Their values of accessibility and inclusivity indicates that a "cool spot" is only viable if it is free, socially welcoming, and physically accessible, such as having step-free entry for strollers or mobility devices.

The Municipality of Amsterdam and the GGD represent the institutional layer of the project, holding the "Data Power" and regulatory authority necessary to sustain the intervention. Their interests are focused on meeting climate adaptation targets and reducing heat-related mortality, but their view is inherently shaped by the need for scalability and institutional efficiency. To them, the intervention must be a "transferable structure" that can be integrated across various municipal departments and maintained over the long term. This institutional logic must be balanced with the more localized interests of private and semi-public entities, such as museums, libraries, and malls. These stakeholders often face a tension between commercial viability and civic duty; while a library may see a cool-seeker as a patron, a shopping mall might perceive them as a non-paying customers. The intervention must navigate these conflicting values by clearly communicating access rules to prevent social friction.

Finally, the CSS Research Team serves as the "Knowledge Broker" within this network. While the team possesses the technical power to synthesize fragmented datasets into a functional tool, this power comes with the ethical responsibility to avoid digital paternalism. This means ensuring the tool is not merely a top-down technical solution but one grounded in the lived experiences and subjective needs of the community. By mediating between the data-heavy requirements of the city and the practical, daily realities of residents, the research team ensures the "Cool Map" remains both technically robust and socially relevant.

4. Theoretical and Practical Framework 1800-2200 words

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Ability to intentionally select appropriate theories and justify their relevance. Students reflect on when and why these theories can be applied to the empirical phenomenon at hand, specifying the conditions under which each theory is applicable and identifying what each one brings into focus or leaves obscured.

Ability to clearly specify the assumptions underlying the selected theories (e.g., regarding human action and social life). As multiple theories are employed, students reflect on how their assumptions relate to one another, for example, whether they are similar, conflicting, or complementary.

Ability to identify the most suitable digital intervention based on a synthesis of insights from academic and policy literature.

- Situate the challenge within existing academic and policy literature
- Identify gaps in current solutions or approaches
- Explain how your digital intervention addresses these gaps
- Justify the chosen theoretical framework with supporting literature
- Formulate research question(s) and hypotheses

5. Data Collection and Analysis 1800-2200 words

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Ability to implement an ethically sound research design and apply social science and data science research methods that are appropriate to the research question(s) and useful for developing an intervention, with convincing justifications for the methodological choices made.

Ability to assess different types of bias in data collection and analysis and to critically reflect on the ethical, legal, and social implications of the research.

- Summarize data collection and analysis methodologies
- Present and interpret key findings
- Explain how findings informed the development of the digital intervention
- Justify ethical considerations in data collection and analysis (refer to appendices)
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Quantitative Data

Qualitative Data

Qualitative data collection combines focus groups and semi-structured interviews. Four focus groups were conducted with elderly residents in heat-vulnerable districts of Amsterdam, specifically Amsterdam Nieuw-West, Amsterdam-Zuidoost, and IJburg. These areas were selected due to their higher exposure to heat-related risks and the presence of populations considered more vulnerable. Focus groups were conducted in Dutch to ensure accessibility, reduce language barriers, and enable participants to express themselves comfortably. Discussions centered on experiences with heatwaves, current coping strategies, and openness to visiting alternative cooling locations.

To complement the focus group data, additional semi-structured interviews were conducted with individuals from a broader range of demographic backgrounds. These interviews allowed for more in-depth exploration of user needs, particularly regarding expectations for a digital map, preferred features, and design considerations. This combination of group-based and individual methods enabled both the identification of shared norms and the exploration of more nuanced, individual-level perspectives.

All data were translated into English prior to analysis. While translation introduces potential risks of meaning loss, care was taken to preserve contextual nuances as much as possible. The data were then analyzed using ATLAS.ti, following a thematic coding approach. An initial coding framework was developed based on the research focus, and iteratively refined as new patterns emerged from the data. **Coding is ongoing; however, several preliminary themes have been identified across both focus groups and interviews.**

Emerging findings indicate multiple forms of resistance to behavioral change. First, some participants reported not experiencing heat as a problem, often as they experience relatively cool indoor living conditions. Second, a group of participants appeared to underestimate heat-related risks, for example by normalizing heat exposure due to prior experience in warmer climates. Third, resistance was linked to negative perceptions of alternative cooling locations, including lack of appeal or the perceived inconvenience of leaving home.

In addition, participants identified several characteristics that influence the attractiveness of cooling locations. These include the presence of engaging or entertaining elements (e.g., opportunities for social interaction or leisure activities), the availability of basic amenities such as toilets, drinking water, and physical accessibility, and the presence of additional “bonus” services such as food or access to medical assistance.

Several forms of bias were considered in both data collection and analysis. Selection bias may have occurred, as participants who chose to take part may differ systematically from those who did not, notably as the focus group participants consisted of elderly people who frequent community centers, elderly people who are more socially distant from society are not represented. Multiple group dynamics may have influenced responses in focus group settings, specifically social desirability bias, conformity and groupthink. While this is less likely in some group sessions as there were many disagreements between participants, in some focus group sessions there was a higher degree of consensus. Translation from Dutch to English introduces the possibility of interpretive bias, as subtle meanings or culturally specific expressions may not be fully preserved. To mitigate these risks, themes were cross-checked across different data sources (focus groups and interviews), and coding decisions were applied consistently throughout the dataset.

Ethical considerations were integrated into all stages of the research process. Participants were informed about the purpose of the study, provided informed consent, and were assured anonymity and confidentiality. They were also informed of their right to withdraw at any time without consequences. Conducting the sessions in participants' native language supported inclusivity and reduced participation barriers. All data were anonymized prior to analysis and handled in accordance with applicable data protection standards. Further details on ethical procedures are provided in the appendices.

6. Digital Intervention 1800-2200 words Alva

Ability to clearly define and prioritize the characteristics of the digital intervention, ensuring alignment with the project's objectives.

Ability to analyse the perspectives of stakeholders, considering their underlying interests, power relations, values, and worldviews for the development of the intervention.

Ability to identify the most suitable digital intervention based on a synthesis of insights from academic and policy literature.

- Outline the key elements of the intervention
- Explain how it addresses the project challenge
- Link the intervention to the theoretical framework
- Show how research findings informed the design
- Discuss relevance and impact for:

- Target audience
- Stakeholders
- Project partner

The digital intervention we develop in this project is called the Amsterdam Cool Map. It is basically a mobile-friendly website that helps people find places to cool down when the city becomes too hot. The main idea sounds simple, but actually it comes from a quite serious problem: Amsterdam already has a lot of climate data and also many places that could help people cool down, but somehow this information is not easy to use in real life.

A heat map, for example, can show where the city is hotter, but it doesn't really help when you are standing outside with your grandma or a baby and wondering "where should I go now?". It doesn't tell you if a place is open, if you need to pay, or even if there is a toilet. So there is this gap between knowing and acting. The Cool Map tries to fill this gap by turning technical data into something more practical, something people can actually use in daily decision.

The map includes different kinds of cooling locations. There are indoor ones, like libraries, community centers, museums, and some semi-public buildings. There are also outdoor ones, like shaded parks, green spaces, and potentially cooler walking areas. But what is more important is not only where these places are, but also how usable they are. From our research, we find that people don't just go somewhere because it is cooler. They care about things like: is it free? is there seating? toilets? drinking water? can I enter with a stroller or wheelchair? If these things are missing, even a "cool place" is not really useful.

This connects to our theoretical framework of vulnerability, which looks at heat risk not just as temperature, but as a combination of exposure, sensitivity, and adaptive capacity. Some people are more exposed, some are more sensitive, but the key difference often comes from adaptive capacity what resources you have to deal with heat. If you have AC at home, maybe you don't need this map. But if you don't, then public spaces become very important. So in a way, this map is also about fairness, or environmental justice, even if we don't always say it directly.

Our qualitative research also influenced the design a lot. Some participants told us they don't really feel heat is a problem, especially if their home is still okay. Others said they don't like to leave home, or they feel other places are boring or inconvenient. And many people mentioned small but important things like toilets, water, somewhere to sit, or just feeling comfortable staying there. Some even said if there is nothing to do, they wouldn't go. So this made us realize: a cooling place is not just about temperature, it is also about feeling welcome, not awkward, not like you don't belong there.

At the same time, working with the GGD was not always smooth (but always productive), there were some disagreements. One main issue is about socio-demographic information. From our

perspective, including this kind of data (like where more elderly people or vulnerable groups live) can help show where heat risk is more serious. It gives a bigger picture, like where the “resilience gap” is. But the GGD is more cautious. They don’t really want this information on the public map, because it can be sensitive or misinterpreted. So in the end, we kind of separate this: for public users, the map stays more simple and practical; for analysis or policy use, this information can still be useful in the background.

Another discussion was about whether to make a phone app. At first, we thought an app could be good, maybe with notifications or real-time features. But the GGD didn’t support this idea. And thinking more about it, it also makes sense apps need to be downloaded, updated, and some people (especially older users) may not want to install something new. So we dropped the app idea and focused on a website instead. A website is easier to access, you just open it, no extra step, which is probably more realistic.

The GGD also suggested adding shade-based route planning, which we actually think is a very good idea. It means not only finding a cool destination, but also walking there in a cooler way, like through shaded streets or parks. This would be especially helpful for elderly people or caregivers. But to be honest, this is quite complex to build, it needs more detailed data and technical work. So for now, we put this as a future feature, not something we can fully achieve in this project.

They also mentioned connecting users with people from similar programs, like some kind of community support. This could help people feel less alone or more willing to go out. We agree this is meaningful, but again, it goes beyond what we can do right now. It would need long-term management and coordination, not just a map. So we also place this idea as something for later development.

For the target audience, which includes elderly residents, caregivers, and other heat-sensitive groups, the Cool Map mainly reduces uncertainty. Instead of searching many places or guessing, they can directly see options and choose what fits them. This may sound small, but in a heatwave situation, it can really matter.

For stakeholders, like libraries, museums, or malls, the map also creates some pressure and opportunity at the same time. They can become part of the city’s cooling system, but it also raises questions: are people allowed to stay without buying anything? will they be welcomed? If this is not clear, it can lead to uncomfortable situations. So the map also indirectly shows the need for clearer agreements.

For the project partner, especially the Municipality of Amsterdam and the GGD, the map has both short-term and long-term value. In the short term, it helps communication during heatwaves. In the long term, it shows where cooling resources are missing. These “gaps” can guide future planning, like where to plant trees or build new facilities.

Overall, the Amsterdam Cool Map is not just a technical tool. It is more like a bridge between data and daily life, between policy and people. And during the process, we also had to adjust

our ideas, negotiate with stakeholders, and accept some limitations. The final result is not perfect, but it is something realistic, usable, and still open for future improvement.

7. Discussion and Conclusion 1800-2200 words - Meng

The primary strength of the Amsterdam Cool Map lies in its ability to translate static environmental data with active social infrastructure. Our findings suggest that heat stress is not merely a failure of temperature management but a failure of information symmetry. By creating a decision-ready tool, the project effectively lowers the search cost for cooling. The overall impact is a shift in the city's adaptation strategy from general awareness to targeted behavioral support. For the broader community, the map serves as a digital commons that validates the lived experience of heat-sensitive residents, transforming the abstract threat of climate change into a navigable urban reality.

For our project partner, the Municipality of Amsterdam, the intervention demonstrates the power of data integration. The project reveals that while the city possesses a wealth of GIS data, it is often siloed. By centralizing libraries, community centers, and "cool routes" into a single mobile-friendly interface, we provide the municipality with a scalable model for future climate interventions.

However, the implications for semi-public stakeholders (like malls or museums) are more complex. The map's success depends on the willingness of these entities to act as climate shelters. If the map directs vulnerable residents to a mall that subsequently discourages non-paying loiterers, it could create social friction. Therefore, the project implies a need for formal "cool spot" agreements between the city and private owners to ensure these spaces are truly inclusive.

Ethically, we must also address the risks of data accuracy and safety. Mapping an indoor space as a cool location creates a social contract with the user. If a resident travels to a designated library during a heatwave only to find the AC broken or the facility closed, the intervention has moved from helpful to potentially hazardous. This highlights a technical limitation: the map is currently dependent on static datasets and community verification rather than real-time temperature sensors or live API feeds from building management systems.

To evolve beyond a prototype, the project should pursue the following three avenues:

1. **Sensor Integration:** Future iterations should aim to integrate real-time temperature and humidity data from indoor locations and sensors in public parks to provide users with real-time comfort levels.
2. **Formal Policy Integration:** The municipality should use the "Resilience Gaps" identified by our spatial analysis to inform future urban planning. The map should not just help people find shade; it should help the city decide where to plant more trees and build new community centers in the neighborhoods where adaptive capacity is lowest.

8. Bibliography

9. Appendices

- Project timeline (including contributor roles by work package)
- Updated stakeholder analysis
- Data collection materials (e.g., consent forms, interview guides, sampling strategies)
- Data and materials for replicating analysis (e.g., code, qualitative data)
- Consent forms and ethical self-evaluation
- Any other relevant documents